**CONFIDENTIAL**




UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Computing

# UNIVERSITI TEKNOLOGI MALAYSIA

## MIDTERM TEST SEMESTER 2, 2017 / 2018

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|---|---|---|
| **SUBJECT CODE** | **:** | **SCSI 1143** |
| **SUBJECT NAME** | **:** | **PROBABILITY & STATISTICAL DATA ANALYSIS** |
| **YEAR/ COURSE/ SECTION** | **:** | **1/ FC/ 01, 02, 03, 04, 05, 06, 07** |
| **TIME** | **:** | **3 HOURS (8:00 – 11:00 PM)** |
| **DATE/DAY** | **:** | **27 MARCH 2018 (TUESDAY)** |
| **VENUES** | **:** | **N28 (BK1-BK6)** |

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**ANSWER ALL (10) QUESTIONS**

| | |
|---|---|
| **Name** | |
| **I/C No.** | |
| **Year / Course** | |
| **Section** | |
| **Lecturer Name** | |

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This question paper consists of **12** printed pages including this page.

**QUESTION 1** **(10 MARKS)**

a) Medical records of four of the patients from a total of 30 patients of diabetes are listed in Table 1.

**Table 1**

| Name | Age | Gender | Blood Sugar |
|---|---|---|---|
| Janet | 30 | Female | 8.5 mol/L |
| Jonas | 35 | Male | 8.0 mol/L |
| Steven | 33 | Male | 7.9 mol/L |
| Zack | 24 | Male | 9.0 mol/L |

According to Table 1, identify the following:

i) Quantitative data [0.5 Mark]
ii) Qualitative data [0.5 Mark]
iii) Sample [0.5 Mark]
iv) Population [0.5 Mark]

b) List and explains the four levels of measurements and provide ONE example for each level of measurements. [2 Marks]

c) Determine the following data whether discrete or continuous.

i) Height of 35 students in a class. [1 Mark]
ii) Number of cars entered the recreation park in weekend. [1 Mark]
iii) The number of fresh graduates from UTM in 2018. [1 Mark]
iv) Average speed of cars passing the federal highway. [1 Mark]

d) List TWO examples of both primary and secondary data sources. [2 Marks]

**QUESTION 2** **(10 MARKS)**

The times in seconds for 15 athletics took to run 800 meters is given Table 2.

**Table 2**

| | | | | |
|---|---|---|---|---|
| 139 | 148 | 151 | 140 | 162 |
| 182 | 154 | 171 | 157 | 142 |
| 145 | 178 | 132 | 148 | 166 |

a) Draw a stem-and-leaf diagram of the data. [3 marks]
b) Find the median, the upper and lower quartiles and the interquartile range (IQR). [4 marks]
c) Draw a boxplot of the data. [3 marks]

**QUESTION 3** **(10 MARKS)**

For the score of golfers in the first round of a competition given in Table 3.

**Table 3**

| Score | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Tally | \| | \|\| | \|\|\|\| | 𝍸 \|\|\| | 𝍸 𝍸 \|\| | 𝍸 \|\| | 𝍸 | 𝍸 \| | | \|\|\| | | \| | \| |

a) Find the median [2 marks]
b) Find the upper and lower quartile [3 marks]
c) Find interquartile range [2 marks]
d) Illustrate the data with boxplot [3 marks]

**QUESTION 4** **(10 MARKS)**

One year the yields, y, of a number of walnut trees were recorded to the nearest kilogram as shown in Table 4.

**Table 4**

| Yield, y (kg) | Frequency | Yield, y (kg) | Frequency |
|---|---|---|---|
| $40 < y \leq 50$ | 1 | $70 < y \leq 80$ | 4 |
| $50 < y \leq 60$ | 5 | $80 < y \leq 90$ | 2 |
| $60 < y \leq 70$ | 7 | $90 < y \leq 100$ | 1 |

a) Construct the cumulative frequency table for the data above. [2 marks]
b) Draw the cumulative frequency graph. [3 marks]
c) Use your graph to estimate the median and interquartile range of the yields. [2 marks]
d) Draw a boxplot to illustrate the data. [3 marks]

**QUESTION 5** **(10 MARKS)**

A random sample of people were asked how old they were when they first met their partner. The histogram in Figure 1 represents this information.

a) What is the modal age group? [1 mark]
b) How many people took part in the survey? [2 marks]
c) Find an estimate for the mean age that a person first met their partner? [2 marks]
d) Draw a cumulative frequency curve for the data. Use the curve to provide an estimate for the median. [5 marks]

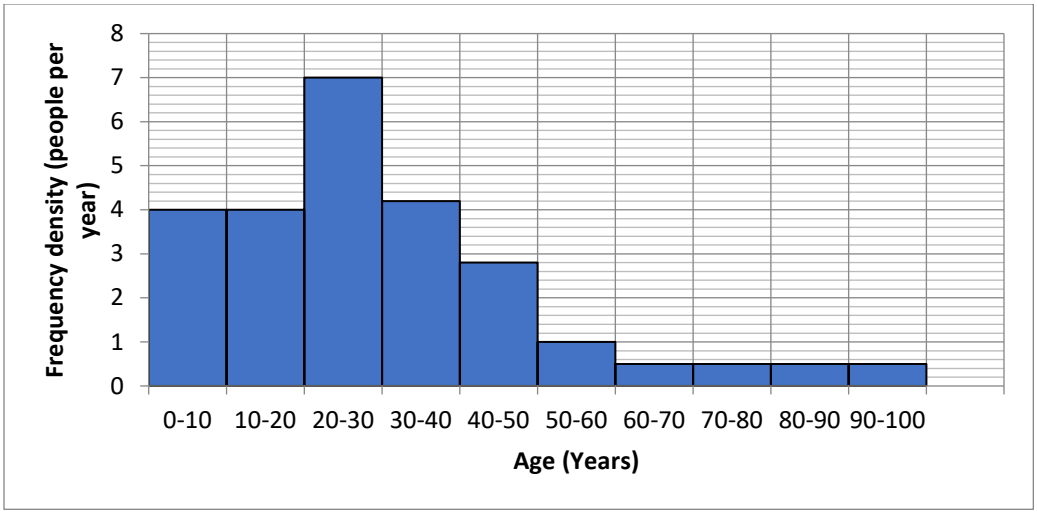



**Figure 1**

**QUESTION 6** **(10 MARKS)**

Table 5 give data on the heights, in cm, of 51 children

**Table 5**

| Class interval | $140 \leq h \leq 150$ | $150 \leq h \leq 160$ | $160 \leq h \leq 170$ | $170 \leq h \leq 180$ |
|---|---|---|---|---|
| Frequency | 6 | 16 | 21 | 8 |

a) Estimate the mean height [3 marks]
b) Estimate the median height [3 marks]
c) Find the modal class [4 marks]

**QUESTION 7** **(10 MARKS)**

Table 6 showing the number of hospitalizations for cardiovascular disease, by age group, for one year. One of these case records is selected at random.

**Table 6**

| **Age group (year)** | **Number** |
|---|---|
| 0-19 | 60 |
| 20-39 | 150 |
| 40-49 | 300 |
| 50-59 | 580 |
| 60-69 | 678 |
| 70-79 | 1288 |
| 80 and over | 1378 |

Find the probability that the patient was

a) In the age of 50s [2 Marks]
b) Less than 50 years old. [2 Marks]
c) Between 40 and 69 years old, inclusive. [3 Marks]
d) 70 years old or older. [3 Marks]

**QUESTION 8** **(10 MARKS)**

In a Japanese restaurant, an average of 3 out of every 5 customers ask for plain water with their meal. A random sample of 10 customers is selected. Let X equals to the number of customers that ask for plain water. Assume each trial is independent.

a) Find the probability that exactly 6 ask for plain water with their meal. [3 Marks]
b) Find the probability that less than 9 customers ask for plain water with their meal? [3 Marks]
c) A second random sample of 30 customers is selected. What is the probability that the first who to ask for plain water is the fourth customer. [4 Marks]

**QUESTION 9** **(10 MARKS)**

Smart phones has been one of the most important part in our daily life. The length of working time for a smart phone before it needs to be charged, $X$ is normally distributed with a mean of 20 hours and a standard deviation of 2 hours.

a) What is the probability that a phone work for at least a day without charging? [3 Marks]
b) Find the value of $X$ given $P(Z) = 0.12$. [3 Marks]
c) You are going to attend a 2-hour meeting right now and your phone has been used for 25 hours since the last charge. What is the probability that your phone will still work throughout the meeting without charging? [4 Marks]

**QUESTION 10** **(10 MARKS)**

A factory mass produces elevator components with 1% defective rate. The components are packed in cartons of 10. A carton is selected at random.

a) Find the probability where:
    i) That box contains only 1 defective component. [3 Marks]
    ii) At least 2 defective components in the box. [3 Marks]
b) Another 5 cartons are selected, find the probability in this batch of sample contains between 1 and 4 (inclusive) defective components. [4 Marks]